

Milankovitch scale (definition): the **long-timescale (about ~20,000 to ~100,000+ years)** climate variation driven by predictable changes in Earth's orbit and tilt—**precession (~23k yrs)**, **obliquity/tilt (~41k yrs)**, and **eccentricity (~100k yrs)**—which change how much sunlight (insolation) different parts of Earth get and can pace ice ages.
Obliquity (tilt): changes the **angle of Earth's tilt** (about 22.1° to 24.5°).

- Bigger tilt → **more extreme seasons** (hotter summers, colder winters), especially at high latitudes.

Precession (wobble): changes the **direction Earth's tilt points** and **when seasons happen** relative to where Earth is in its orbit (closest vs farthest from the Sun).

- It mainly shifts **which hemisphere gets stronger seasons**.

Eccentricity (orbit shape): changes how **circular vs elliptical** Earth's orbit is.

- More elliptical → bigger difference between **perihelion (closest)** and **aphelion (farthest)**, which makes precession's seasonal effect **stronger**.

Molecules & Life Central Dogma (flow of information) Correct sequence (example: HBB → hemoglobin): HBB DNA → HBB mRNA → beta-globin amino acid sequence → beta-globin 3D shape → hemoglobin assembly → hemoglobin function ML1_HW_ANSWERS_Fall2025 (4) Transcription: DNA → mRNA using RNA polymerase (DNA damage can make wrong mRNA) Practice_Final_ANSWERS_Fall2025... Translation: mRNA → protein using ribosome (bad mRNA → bad protein) Practice_Final_ANSWERS_Fall2025... Ribosome facts: does translation; made of RNA + proteins Practice_Final_ANSWERS_Fall2025... Nucleotides, codons, amino acids (the "3" and the "4") Nucleotides = letters DNA letters: A, T, C, G (4 choices) RNA letters: A, U, C, G (4 choices) Codon = a 3-letter "word" (exactly 3 nucleotides). 1 codon → 1 amino acid (protein "building block"). Example: AUG is one codon (A+U+G) → one amino acid (also the common "start" codon). Cancer (molecular basis) Cancer = cells grow/divide uncontrollably because mutations hit key growth-control genes: Tumor suppressors = "brakes" (stop growth when something's wrong) Proto-oncogenes = "gas pedal" (promote growth) DNA damage (radiation/toxins) or inherited mutations can cause this; if DNA repair fails, mutations persist → cancer Practice_Final_ANSWERS_Fall2025... Mutations + sickle cell specifics Healthy vs mutated protein segment: Glu → Val causes hemoglobin to clump (sickling) ML1_HW_ANSWERS_Fall2025 (4) Codon chart use (how to read): first 2 bases set the "box," third base picks the amino acid; STOP codons exist ML1_HW_ANSWERS_Fall2025 (4) "Which machine caused it?" Replication error: DNA polymerase (can be inherited) ML1_HW_ANSWERS_Fall2025 (4) Transcription error: RNA polymerase ML1_HW_ANSWERS_Fall2025 (4) Translation error: ribosome ML1_HW_ANSWERS_Fall2025 (4) Treatment idea: drug binds mutated Val to reduce polymer formation/sickling ML1_HW_ANSWERS_Fall2025 (4) Structure methods: X-ray crystallography (X-rays) and Cryo-EM (electron beams) ML1_HW_ANSWERS_Fall2025 (4) Probability + "at least once" (super important) Core rules (your notes) Probability = (# favorable outcomes)/(# equally likely outcomes) (your note) Independent AND: $P(A \text{ and } B) = P(A)P(B)$ OR (mutually exclusive): $P(A \text{ or } B) = P(A) + P(B)$ At least once: $P(\geq 1) = 1 - P(\text{none}) = 1 - (1 - p)^n$ This is exactly what you used in several places (and what your HW key uses too) ML1_HW_ANSWERS_Fall2025 (3) "Choices^positions" counting rule (your note) If you have k choices for each of n slots, total outcomes = k^n . (Example you had: 4 options each time for 6 slots → 4^6 .) FoS Birthday practice questions (from the tutorial sheet) Questions 1–8 are explicitly listed here Tutorial_Practice_Questions_FIN... Your "birthday-style" answers should use: One specific day (e.g., July 4): 1/365 OR of two specific days (July 4 OR Feb 14): 2/365 (assuming non-leap year, mutually exclusive) AND of two different birthdays for the same person: 0 (impossible) Tutorial_Practice_Questions_FIN... Same birthday as you: 1/365 Same birthday AND female: $(1/365) \cdot (1/2)$ At least one match in a group: if there are n other students, $1 - (364/365)^n$ DNA damage probabilities (ML1 HW #2) Given: Genome length ≈ 3.1 billion nucleotides ML1_HW_ANSWERS_Fall2025 (4) Hemoglobin coding region for 140 amino acids: $140 \times 3 = 420$ nucleotides ML1_HW_ANSWERS_Fall2025 (4) Daily DNA damage events: 50,000/day ML1_HW_ANSWERS_Fall2025 (4) Key results: Odds one mutation hits the HBB coding region: $420 / (3.1 \times 10^9) = 1.35 \times 10^{-7}$ ML1_HW_ANSWERS_Fall2025 (4) Probability none of 50,000 hits HBB: $(1 - 1.35 \times 10^{-7})^{50000} = 0.993$ ML1_HW_ANSWERS_Fall2025 (4) Probability at least one per day hits HBB: $1 - 0.993 = 7 \times 10^{-3}$ ML1_HW_ANSWERS_Fall2025 (4) Probability at least one over a year (this answers your "0.993^365 vs 1 - 0.993^365" confusion): $0.993^{365} = \text{"NO hit all year"}; \text{Answer wanted} = \text{"}\geq 1 \text{ hit all year"}: 1 - (1 - 7 \times 10^{-3})^{365} = 0.923$ ML1_HW_ANSWERS_Fall2025 (3) Protein folding toy model (ML1 HW #1B) (i) Brute force folding time Assumptions: 140 amino acids, 4 configs each, 0.5 ns per config ML1_HW_ANSWERS_Fall2025 (4) Total states: $4^{140} = 1.94 \times 10^{84}$ ML1_HW_ANSWERS_Fall2025 (4) Time in Gyr: $\sim 3.07 \times 10^{58}$ Giga-years

ML1_HW_ANSWERS_Fall2025 (4) (Shows why proteins must fold smarter than brute force.) (ii) Faster “secondary structure first” model 10 aa per alpha helix; 140 aa → 14 helices ML1_HW_ANSWERS_Fall2025 (3) Each helix: 4^{10} states; they fold in parallel ML1_HW_ANSWERS_Fall2025 (3) Subunits: 4 subunits total; “limiting” one has 4 helices, each helix has 6 configs → $6^4=1296$ ML1_HW_ANSWERS_Fall2025 (3) Final folding steps: 20 steps × 0.5 ns ML1_HW_ANSWERS_Fall2025 (3) Total folding time $\approx 5.250 \times 10^{-4}$ seconds ML1_HW_ANSWERS_Fall2025 (3) Parallel-folding rule (the thing you were stuck on): If parts fold “at the same time,” you usually don’t multiply times by the number of parts—you take the slowest (limiting) time, then add the sequential stages. Clinical trials + statistics interpretation (Practice Final) Control groups (why compare to existing drugs) Control group is needed to show new drug is better than existing; placebo-only can be unethical in lethal disease Practice_Final_ANSWERS_Fall2025... Hazard Ratio (HR) cheat HR < 1 → experimental (selpercatinib) better; HR = 1 → no difference; HR > 1 → control better; “Significant?” → 95% CI doesn’t cross 1 (or p-value very small) Selpercatinib conclusion: Selpercatinib appears more effective; HR is significant ($p < 0.001$) Practice_Final_ANSWERS_Fall2025... Curves become significantly different ~8–9 months when 95% CIs stop overlapping Practice_Final_ANSWERS_Fall2025... ~80% progression-free at 33 months Practice_Final_ANSWERS_Fall2025... Subgroups: No significant age-group difference if CIs overlap Practice_Final_ANSWERS_Fall2025... Cabozantinib vs vandetanib: overlap → can’t say one better; cabozantinib worse than selpercatinib; vandetanib ~ no better/worse if CI overlaps HR=1 Practice_Final_ANSWERS_Fall2025... Mechanism of drug (molecular therapeutics idea): Selpercatinib = small molecule inhibitor binding RET kinase to stop growth signaling Practice_Final_ANSWERS_Fall2025... IRR (incidence rate ratio) cheat (malaria + sickle cell) IRR = 1 → equal risk; Significant protection when 95% CI does not overlap 1 Practice_Final_ANSWERS_Fall2025... In older kids, overlap → not significant Practice_Final_ANSWERS_Fall2025... CRISPR (what it is + how it was discovered) In the Practice Final, CRISPR-Cas9 is treated as a genome editing method; probability-of-success work uses 0.9^{16} and then $(0.185)^{100}$ logic Practice_Final_ANSWERS_Fall2025... Key probability chain: All 16 edits in one cell: $0.9^{16}=0.185$; All 100 cells: $0.185^{100}=5.21 \times 10^{-74}$ Practice_Final_ANSWERS_Fall2025... Therapy comment: may be impractical if you truly need “all/most cells perfectly edited,” but could still help if partial editing reduces risk Practice_Final_ANSWERS_Fall2025... (From your notes request “how CRISPR was discovered”): CRISPR was first noticed as repeating DNA patterns in bacteria/archaea; later understood as a bacterial immune system against viruses; Cas9 is a nuclease guided by RNA to cut matching DNA; cells repair the cut, which scientists hijack for editing. Blood transfusion calculation (Practice Final 2C(ii)) Given: Each bag: 180 mL RBCs, and RBCs are 55% of bag volume Practice_Final_ANSWERS_Fall2025... +1 g/dL Hb needs 0.01 bags/kg Practice_Final_ANSWERS_Fall2025... Need raise 6 → 10: $\Delta = 4$ g/dL Steps (matches what you wrote): Bags needed = $(10-6) \times 0.01 \times (\text{body weight in kg})$; Total volume per bag = 180/0.55 mL; Donor blood volume = (bags) × (volume per bag); Your method “180/0.55 then multiply by 70×0.04” is the same structure. Stats toolbox (from your notes + practice style) Standard deviation (sample SD) Mean: $\bar{x}=(\sum x_i)/n$; Deviations: $x_i - \bar{x}$; SS: $\sum (x_i - \bar{x})^2$; Sample variance: $s^2=SS/(n-1)$; SD: $s=\sqrt{s^2}$ Standard error and confidence intervals $SE \approx SD/\sqrt{n}$ (used heavily in “error bars” reasoning); 95% CI $\approx \text{mean} \pm 2 \times SE$ Log plots + scaling (Tutorial practice) Be able to label ticks and read values on log-log plots and semi-log plots Tutorial_Practice_Questions_FIN... Be able to extract coordinates of labeled points on a log plot (table A–K task) Tutorial_Practice_Questions_FIN... Be able to plot stride frequency vs mass (linear + semi-log) using Heglund et al. dataset Tutorial_Practice_Questions_FIN... Orders of magnitude practice (Tutorial) Know how to compare sizes as powers of 10 (person vs proton, Moon distance vs skyscraper, etc.) Tutorial_Practice_Questions_FIN... Quick “reference formulas / constants” (Practice Final cover sheet) Units: 1 km= 10^3 m, 1 m= 10^2 cm, 1 kg= 10^3 g, 1 m³= 10^3 L Practice_Final_ANSWERS_Fall2025... Geometry: $A_{\text{circle}}=\pi r^2$; $SA_{\text{sphere}}=4\pi r^2$; $V_{\text{sphere}}=(4/3)\pi r^3$; $V_{\text{cyl}}=h\pi r^2$ Practice_Final_ANSWERS_Fall2025... Motion: $d=vt$ Practice_Final_ANSWERS_Fall2025... Density: $D=m/V$ Practice_Final_ANSWERS_Fall2025... Climate/earth bits from your screenshots (keep these) Milankovitch cycles (your cheat lines) Eccentricity (~100k yr): orbit shape (more circular ↔ more oval) → changes seasonal/orbital contrast; Obliquity/tilt (~41k yr): tilt angle → stronger seasons with more tilt; milder summers with less tilt (ice persists); Precession (~23k yr): “wobble” changes which season occurs at perihelion/aphelion → shifts sunlight timing; Memory: Shape (100k) + Tilt (41k) + Wobble (23k) → sunlight distribution → ice ages Climate proxies (your list) Tree rings: annual; temp/precip; not 800k years (max ~12,400y); Ice cores: layered ice; trapped gases → past CO₂/temp; can cover hundreds of thousands of years; Ocean sediment cores: shells/chemistry; past ocean temp/ice volume; can reach millions of years; Speleothems: rainfall/temp; usually thousands of years; Corals: decades–centuries; Fossils/rocks: very long timescales (even billions) Carbon isotopes (your note) If ¹²C/¹³C goes up → ¹³C/¹²C goes down, and scientists often say $\delta^{13}\text{C}$ decreases (more negative). 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